

MID-YOLO: An enhanced YOLOv8-based method for multi-type insulator defect detection

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HIGHLIGHTS

- A multi-type insulator fault detection strategy MID-YOLO is proposed for detecting more subtle and challenging flashover problems.
- The network neck structure is redesigned based on YOLOv8 to enhance the network's ability to distinguish features in complex contexts.
- We conceptualize and propose a detail-aware module that enhances the model's sensitivity to small-scale targets by restructuring features and integrating them with attention mechanisms.
- A novel multi-scale residual block is created, which not only improves the model's ability to perceive complex details but also significantly reduces computational requirements.

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ABSTRACT

Insulators in overhead transmission systems are susceptible to the adverse effects of harsh weather and other conditions in outdoor environments, posing threats to the safety and stability of the transmission systems. Currently, deep learning-based object detection algorithms have made significant strides in the field of insulator defect detection. However, existing methods mainly tackled insulator broken, overlooking the nuanced detection of complex flashover issues, which require a robust detection network, this research advances insulator defect detection by proposing a novel algorithm based on the YOLOv8 framework, aimed at accurately localizing diverse insulator faults within complex natural scenes. Key innovations include a new neck feature fusion structure and a detail-aware module, which together enhance the model's proficiency in identifying small defects. These modifications have led to notable improvements: a 3.1% increase in average accuracy and an 8.2% uplift in recall rate. Specifically, the CFM module's efficient fusion and TCB's channel weighting significantly boost feature extraction, the MRB's multi-branch strategy increases recall while reducing parameters, and the DAB's attention mechanism sharpens detail perception. The developed MID-YOLO model matches the performance of more parameter-heavy counterparts, offering a practical solution for insulator defect detection.

1. Introduction

Insulators play a crucial role in maintaining the safety and stability of power transmission within the electrical system. They effectively isolate metal structures such as wires and supports, uphold the transmission lines within the power system, and prevent short-circuiting and leakage, thereby ensuring stable power transmission. Despite being designed to withstand natural disturbances like lightning strikes and heat, transmission lines, being largely exposed to the natural environment, face safety challenges due to the complex geography and natural conditions.

Prolonged exposure to environmental disturbances inevitably leads to the aging and damage of insulators, further leading to transmission failures. Taking porcelain insulators as an example, minor cracks on the insulator head can allow external moisture to penetrate, reducing the head's insulation resistance. In such cases, lightning strikes or similar events can cause flashovers, where the heat generated by the electric arc can lead to insulator explosions and performance degradation, potentially causing power outages. Therefore, it is essential to regularly inspect power insulators.

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In recent years, with the advancement of drone technology and image processing, the method of using drones for electrical inspections has gradually replaced manual inspections, making the inspection process more efficient and safe. Nowadays, power transmission systems are distributed everywhere, including forests, rivers, and lakes, which complicates the backgrounds of images captured by drones. Detecting insulator defects remains a challenging task due to the small size of the defects and the heavy background clutter. Traditional computer vision methods [2,4,21] typically consider using color, shape, texture, and contour as features for detecting insulator defects. Zhai et al. [26] proposed an insulator defect detection algorithm based on saliency and adaptive morphology, which uses color and gradient features to locate the position of insulators and separates the insulators from the scene using a color model, followed by defect detection and localization through adaptive shape analysis. However, traditional methods lack robustness and struggle to adapt to various complex backgrounds, resulting in lower reliability and failing to meet the accuracy requirements of electrical inspections.

Compared to traditional image processing algorithms, convolutional neural networks (CNNs) have the ability to learn high-level image features, thereby breaking the deadlock in object detection algorithms and developing at an unprecedented pace. In 2014, Girshick et al. [5] pioneered the CNN region-based algorithm R-CNN, which achieved significant progress on the VOC07 dataset, improving the mean accuracy by 24.8%. However, its drawbacks were also apparent, such as the overlap in candidate boxes leading to a large amount of redundant computation and consequently, very slow detection speeds. To address this, He et al. [7] introduced SPPNet, incorporating a spatial pyramid pooling layer that allows CNNs to generate fixed-size feature blocks without the need to repeatedly resize images, thus avoiding redundant convolutional computations and improving the speed by 20 times compared to R-CNN. Despite these advancements, both methods utilized a two-stage, coarse-to-fine detection approach, which was structurally complex and difficult to speed up, making them challenging to apply in engineering projects. Therefore, Redmon et al. [18] proposed You Only Look Once (YOLO), which divides the image into multiple regions and simultaneously predicts the bounding boxes and probabilities for each region. On the VOC07 dataset, YOLO achieved an mAP of 52.7% while reaching speeds of 155 frames per second. Chen et al. [1] employed an attention feedback mechanism, which helped the model to focus more on the critical regions of insulator defects and improved the accuracy of detection. The dual-space pyramid structure is used to capture feature information at different scales, which enhances the model's adaptability to complex backgrounds and target changes. However, the complex attention mechanism and dual-space pyramid structure may increase the computational complexity and training time of the model. The development of deep learning has laid the foundation for the intelligent construction of power systems. Algorithm design for power inspection has made significant progress across various electrical scenarios.

Liu et al. [16] addresses the current research difficulties of small-scale objects, complex backgrounds, and the limited amount of collected data, and provides an overview of the insulator processing stage and defect detection stage based on deep learning methods according to different processing stages. Detection tasks in different power scenarios each have their unique characteristics, making it difficult for existing object detection models to be directly applied. Consequently, many studies [14,17,30] have further optimized models based on their specific features. Due to the small size of insulator defects and their distribution across various complex backgrounds with low distinguishability, the detection accuracy of general models decreases. To achieve better detection performance, improvements must be made considering these characteristics. Tao et al. [20] introduced a new two-stage detection method into the task of detecting insulator defects. They employed data augmentation techniques such as Gaussian blur and affine transformations to enhance the model's generalization capability, resulting

in an increase in model accuracy from 0.91 to 0.96. However, this method incurs significant training and computational costs, making it difficult to apply. To address this, Zhang et al. [27], building on YOLOv3 technology and by incorporating an FPN structure [13], enhanced the network's fault detection capability while reducing the risk of overfitting. Following this, Yang et al. [25] further increased detection speed and effectively improved the detection accuracy of small objects such as insulators by modifying the algorithm's loss [28] function and feature connection structure. Zhou et al. [31] combined the lightweight architecture of YOLOv5 [11] with the capabilities of a two-stage detector to enhance the detection of small objects. Furthermore, by incorporating rotated bounding boxes [3], they improved the quality of the predicted boxes, achieving an accuracy of 98%. However, the adoption of a two-stage strategy also significantly reduced the frame rate.

In a similar vein of innovation, Liu et al. [15] proposed a weather-domain synthesis network for extracting cross-modal discriminative information in a multi-domain insulator defect detection and classification task. Weather-domain migration learning is incorporated to enable the model to maintain stable detection performance under different weather conditions. The accuracy of detecting small targets is improved by adding a target detection layer to fuse shallow and deep feature mappings. Zhang et al. [29] improved the feature extraction block in YOLOv5 using attention mechanisms and enhanced the dataset's size and complexity through fog data augmentation. This effectively improved the model's detection capability across various scenarios, increasing the mAP50:95 by 1.3%. However, these methods often only optimize one aspect, either the network structure or the feature extraction mechanism, resulting in only modest performance improvements. Acknowledging this gap, He et al. [8] set out to refine the model's capacity to differentiate complex backgrounds. By integrating GhostNet [6] with multi-scale asymmetric convolution, they upgraded the C2f network. Their design of a new connection structure substantially amplified the model's ability to detect targets of varying sizes. This strategic overhaul not only slashed the model's computational demand but also elevated the detection accuracy from 89.2% to an impressive 93.9%. Their approach underscores the necessity of tailoring detection strategies to the nuanced differences across feature types and levels within the network. However, it did not account for the differences between features across various parts of the network, leading to redundant module features and limiting efficiency.

Previous methodologies predominantly focused on addressing issues related to insulator explosions and subsequent losses, while the detection of subtler and more challenging flashover issues necessitated a more robust detection network. In response to this need, our research introduces a multiple types of insulator fault detection strategy MID-YOLO, highlighted by the following key contributions:

1. Leveraging the foundation of the YOLOv8 [12], we reengineered the neck structure to bolster the network's ability to distinguish features against intricate backgrounds. This was achieved by transitioning the fusion component to a Trainable Concat Block, optimizing the efficiency of feature fusion.
2. Acknowledging the discrepancy between the characteristics of neck features and those of the main network, we transitioned from a depth-focused architecture to a breadth-oriented structure. This led to the creation of an entirely new Multi-Scale Residual Block, which not only elevates the model's ability to perceive intricate details but also substantially reduces computational demands.
3. Through an in-depth analysis of the network's neck output features, our research identified an underutilization of repetitive features. To address this, we conceptualized and implemented the Detail Aware Block, a pioneering approach that enhances the model's sensitivity to small-scale targets by reorganizing features and integrating them with an attention mechanism.

These contributions collectively underline our commitment topushing the boundaries of insulator fault detection, offering a robust solution capable of navigating the complexities of real-world applications.

2. Proposed method

2.1. Overview

The cutting-edge YOLOv8 model was selected as the foundational framework for this study. At its core, a gradient flow-enriched C2f structure is utilized, and an Anchor-Free architecture for the detection head has been adopted, further enhanced by the introduction of Distribution Focal Loss. These improvements have significantly enhanced the model's performance, however, they have also introduced an increased computational load and network parameters. Furthermore, challenges such as insulator defects and flashover faults occupy a minor fraction of the overall image, highlighting the need for improved detection of small targets by YOLOv8. To address this, the more efficient MID-YOLO structure is proposed, built upon the YOLOv8 framework. The structure of MID-YOLO is illustrated in Fig. 1. The MID-YOLO incorporates an efficacious CFM neck feature fusion structure and introduces two specially designed modules: a lightweight MRB, employing a multi-branch breadth design to replace the traditional deep design strategy, thus enhancing the network's perceptual capabilities; and a simplified yet effective TCB, which leverages channel weighting to augment the network's spatial dependency awareness. Lastly, by integrating a DAB before the detection head, the refined features processed by the neck are enhanced with positional information, long-range relationships, and global feature enhancement, effectively improving the model's performance in detecting small targets. A detailed discussion of each component is provided in the following sections.

2.2. Concordance fusion module

After extraction by the network backbone, the multi-level features contain diverse semantic information. Given this, the neck of the network requires a stronger capability for feature integration rather than merely enhanced feature extraction. This paper introduces the Concordance Fusion Module (CFM), based on the BiFPN neck structure [19], to address this need. This module features an increased number of bidirectional connections, enabling the fusion of more features at the same cost. The CFM network structure is illustrated in Fig. 2, this approach alters the feature fusion paths within the YOLOv8 network

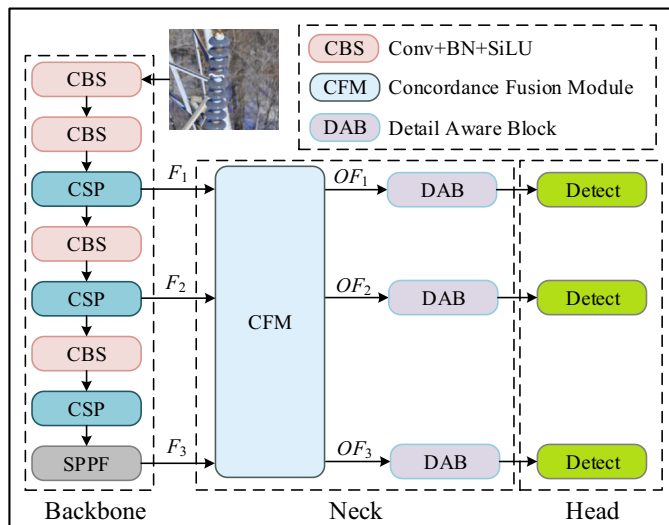


Fig. 1. Structure of MID-YOLO.

framework, utilizes a Trainable Concat Block (TCB) to differentiate the weights of various features, and ultimately replaces the original C2f block with a more efficient Multi-Scale Residual Block (MRB).

The feature fusion strategy of CFM consists of two stages. In the bottom-up stage, the feature maps F_1 , F_2 , and F_3 have dimensions of 160×160 , 80×80 , and 40×40 , respectively. Initially, F_3 is processed through a CBS block to obtain F'_3 , which is then upsampled to match the size of F_2 's feature map. This is followed by fusion with F_2 through a TCB, and the fused feature map is further refined by an MRB to yield F'_2 . Repeating these steps leads to the generation of the top-layer feature F'_1 . The top-down process is similar, except that the CBS uses a convolution kernel with a stride of 2 for feature extraction and down-sampling. During the creation of F'_2 , the TCB block receives inputs from F'_1 , F_2 , and F'_2 , allowing the CFM computation to be represented as follows:

$$W_1 = f_{\text{conv}}^{1 \times 1}(F) \quad (1)$$

$$W_2 = f_{\text{dconv}}^{3 \times 3}(W_1) \quad (2)$$

$$W_3 = f_{\text{conv}}^{1 \times 1}(W_1) \quad (3)$$

$$W_4 = f_{\text{conv}}^{1 \times 1}(W_3) \quad (4)$$

$$O = f_{\text{conv}}^{1 \times 1}(\text{Concat}(W_1, W_2, W_3, W_4)) \quad (5)$$

where $f_{\text{conv}}^{3 \times 3}$ and $f_{\text{conv}}^{1 \times 1}$ denote convolution operations with kernel sizes of 3×3 and 1×1 , respectively, while $f_{\text{dconv}}^{3 \times 3}$ refers to a dilated convolution [22] operation with a dilation rate of 2. 'Concat' signifies an operation for concatenating features, F represents the input feature to the MRB module, and O is the output feature from the MRB module. Compared to the C2f module, the MRB module features a simpler structure and employs a multi-branch extraction and fusion strategy, enabling the model to learn more complex contextual features and enhance its ability to represent small targets, such as insulator flashover faults.

The evolution of attention mechanisms has demonstrated that different channels of features contribute variably to network performance. The original YOLOv8 used basic feature concatenation, which did not fully exploit the potential of features across different channels. To achieve more efficient multi-scale feature fusion, incorporating a channel attention mechanism is deemed feasible but would significantly increase the model's computational and parameter load. Consequently, this paper introduces a more efficient, weight-trainable connective block (TCB). Inspired by fully connected structures, trainable parameters equal to the number of feature channels are added after the Concat operation to weight different channels. The calculation process is shown as follows:

$$\text{Output} = \sum_{i=1}^N \frac{\omega_i}{\epsilon + \sum_{j=1}^N \omega_j} \cdot x_i \quad (6)$$

Where N represents the total number of feature layers after concatenation, x_i represents the feature from the i th layer, and ω_i is the weight associated with the feature from the i th layer. The calculation of $\sum_{j=1}^N \omega_j$ is intended for the normalization of weights, and ϵ is set to 0.0001 to prevent division by zero errors. Compared to channel attention mechanisms, this method does not significantly increase the number of network parameters.

The CFM module enhances the network's fusion process for multi-scale features through a more efficient connection structure and feature extraction blocks. This allows for finer extraction of small target features, laying the groundwork for improved detail perception in subsequent sections.

2.3. Detail aware block

Following the multi-scale feature fusion through the CFM, the network is now adept at representing target features. At this stage, a global

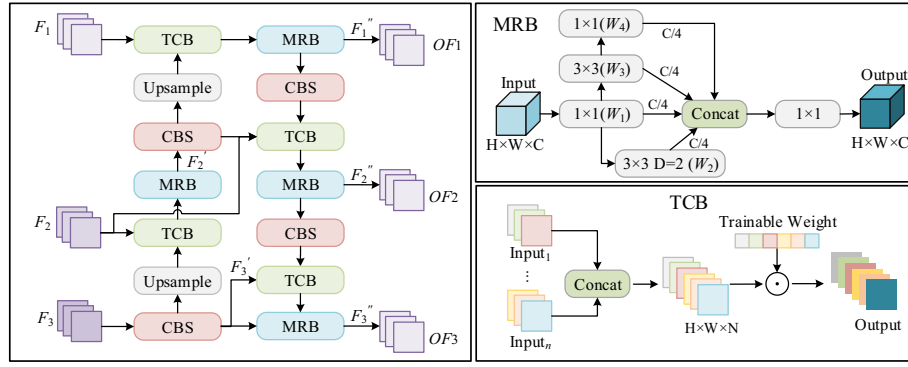


Fig. 2. Structure of CFM.

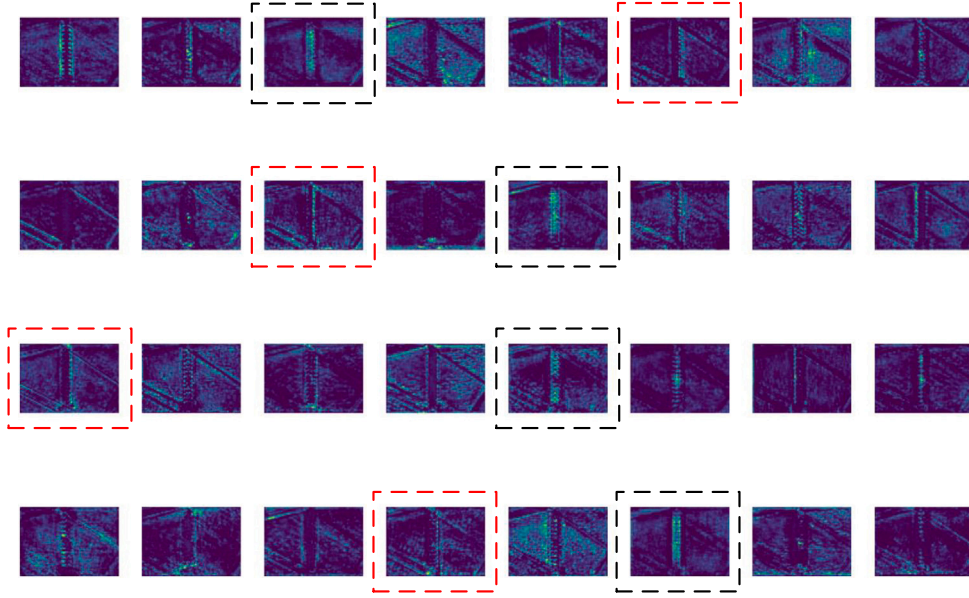


Fig. 3. Visualization of CFM output features with red and black boxes indicating two groups of similar features.

perception of different feature maps is utilized, making the use of global contextual information for deep modeling of small target details more effective than in previous stages. The CA (Coordinate Attention) [9] mechanism differs from widely used methods such as SENet [10] and CBAM [23] by considering both channel and spatial dimensions. These dimensions are independently encoded, allowing this strategy to capture remote dependencies in the spatial direction while retaining precise location information. Given that defects such as flashover are minor in the overall image, with their location and feature information not being prominent, and considering that Fig. 3 displays CFM output features with many similar features, simply employing CA to encode the entire image is inefficient and unnecessary. Thus, this paper designs the Detail Aware Block (DAB), as shown in Fig. 4. With numerous similar features present in the output, DAB initially utilizes a 1×1 convolution for feature rearrangement. Subsequently, the rearranged features are divided into two groups, significantly reducing the computational load of the module. The computational flow of DAB can be represented as follows:

$$Y = f_{\text{conv}}^{1 \times 1}(IF) \quad (7)$$

$$S_1 = f_{\text{dconv}}^{3 \times 3}(Y') \quad (8)$$

$$S_2 = f_{\text{CA}}(Y'') \quad (9)$$

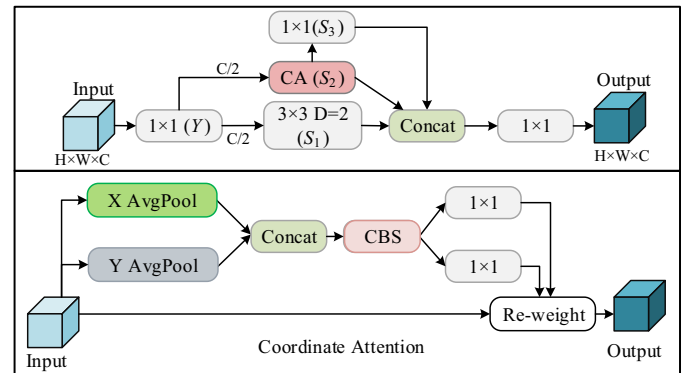


Fig. 4. Structure of DAB.

$$S_3 = f_{\text{conv}}^{1 \times 1}(S_2) \quad (10)$$

$$EF = f_{\text{conv}}^{1 \times 1}[\text{Concat}(S_1, S_2, S_3)] \quad (11)$$

where f_{CA} represents the computation within the CA block, IF is the input feature to the DAB, and EF is the output feature enhanced by the DAB. Y' and Y'' denote the features obtained by equally splitting Y along the channel dimension. Y' undergoes a dilated convolution with a broader receptive field to enhance the DAB's perception of global features, resulting in the intermediate feature S_1 . Y'' is processed through the CA block for more efficient feature handling, yielding the intermediate feature S_2 . Subsequently, a 1×1 convolution adjusts the channels and fuses the feature information to produce feature S_3 . By merging the features from the three branches, the DAB module generates an enhanced feature map with the same scale but integrated with precise location information, long-range relationships, and global features, effectively improving the model's detection performance for small targets.

3. Experience

3.1. Datasets description and experimental environment

Considering the model's practicality, this study employs the FBID dataset [24] for research. This dataset encompasses three types of data: "insulator", "broken", and "pollution_flashover". The distribution of various detection targets within the dataset is as indicated in Table 1. The dataset exceeds the average number of samples in the VOC2012 standard dataset, 1,154, thus providing sufficient data to support neural network learning. As demonstrated in Fig. 5, the dataset includes a substantial number of small-pixel targets, with those having pixel dimensions less than 80×80 making up 60.5% of the dataset. Therefore, given its extensive inclusion of small-pixel targets, the FBID dataset is particularly suited for validating the model's performance in detecting small objects, positioning it as an ideal benchmark dataset for insulator defect detection.

The model is implemented on the PyTorch platform, with all tasks deployed on an NVIDIA 3090 GPU. The training is set for 300 epochs with a batch size of 16 and an initial learning rate of 0.01. The training process is optimized using the SGD optimizer. To reflect the model's detection performance on small targets, mAP is used as the primary evaluation metric.

3.2. Ablation experimental result

To analyze the importance of each component within MID-YOLO, this section uses YOLOv8n as the baseline model and incrementally adds BiFPN, TCB, MRB, and DAB to verify their effectiveness. Fig. 6 displays the loss curves during the training process with different components.

Table 1

Distribution of FBID dataset.

Target	insulator	broken	pollution_flashover
Train	1480	901	1941
Test	352	192	507

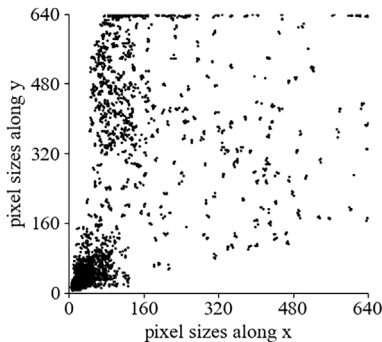


Fig. 5. Sample Size Distribution in the Dataset.

It is evident that with the inclusion of MID-YOLO's specially designed modules, the model's convergence speed, loss, and accuracy have all shown significant improvements.

To analyze the specific contributions of each component, the performance of the model on the FBID test set after sequentially adding each component is compared in Table 2. Evidently, compared to YOLOv8n, MID-YOLO shows significant improvements across all metrics. Specifically:

- (1) BiFPN: Although MID-YOLO incorporates the BiFPN neck structure, the improvement after changing the connection structure is not pronounced, with mAP50:95 only increasing by 0.6, and a slight decrease in precision. This indicates that merely swapping the neck structure without targeted design does not effectively enhance network performance.
- (2) TCP: The results demonstrate that further adding TCP to the network significantly improves all evaluation metrics, especially Precision (+1.5) and mAP50:95 (+2.6), while only increasing the parameter count by 0.12M. This proves that the strategy of weighting channels through TCP can efficiently utilize features across different channels, thereby enhancing the network's feature extraction capability.
- (3) MRB: Incorporating the MRB module not only reduces the network's parameter count by 0.16M but also further improves all metrics, particularly the Recall (+6.7). This proves that abandoning the original, more complex and redundant depth feature extraction strategy in the neck part for a multi-branch, low-dimensional breadth feature extraction strategy is more efficient.
- (4) DAB: The experimental results show that adding DAB further improves all metrics, proving that performing detail-aware operations after neck feature extraction enables the network to more easily detect small objects. To further validate this conclusion, processed visualization feature maps are displayed in Fig. 7. Brighter areas indicate the network's focal points, showing that the CFM neck can roughly filter feature ranges, while DAB allows for more precise localization of small target positions, thereby comprehensively enhancing network performance. Thanks to DAB's feature rearrangement and grouping design, even with the addition of an attention mechanism, the model's parameter count remains within an acceptable range.

3.3. Analysis of the model's robustness under different conditions

Fig. 8 presents a comparison of detection results between MID-YOLO and YOLOv8 across various scenarios. In (a) and (b), medium-sized flashover defects are included, but YOLOv8 failed to detect them in both cases, whereas MID-YOLO effectively identified all defects. In scenarios (c) and (d), which feature many small-pixel defects, YOLOv8 experienced more severe missed detections, while MID-YOLO successfully addressed this issue, proving the effectiveness of the targeted optimizations in this study. (e) and (f) show cases of broken defects, with MID-YOLO demonstrating better accuracy and confidence compared to YOLOv8. In (g), YOLOv8 incorrectly identified a damaged insulator set as two separate sets, and in (h), YOLOv8 missed both the insulator set and broken defects, demonstrating MID-YOLO's superior robustness across different target types and sizes.

3.4. Comparisons with previous methods

To demonstrate the superior performance of the MID-YOLO model, comparisons were made with existing studies, including general lightweight models like YOLOv5n, high-performance models such as SparseRCNN, and models designed for insulator defect detection like FINet and MFI-YOLO. Table 3 shows the AP values of different models for various target types. It is evident that most models can accurately identify insulators, but have lower detection accuracy for Broken and

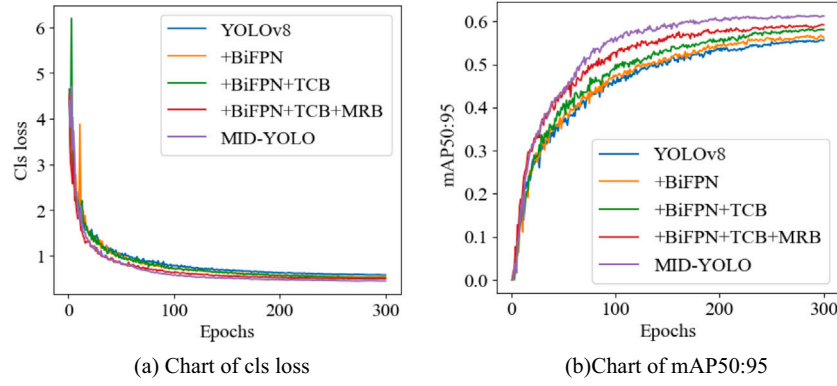


Fig. 6. The impact of various model improvements on the training process, with (a) showing loss changes and (b) showing mAP50:95 changes.

Table 2

Ablation experiments for BiFPN, CFM, TCP, MRB, DAB.

BiFPN	TCP	MRB	DAB	P (%)	R (%)	mAP50 (%)	mAP50:95 (%)	Para (M)
×	×	×	×	90.8	80.4	87.1	55.6	2.15
✓	×	×	×	89.7	83.0	87.1	56.2(+0.6)	2.17
✓	✓	×	×	92.3	83.1	87.5	58.2(+2.6)	2.29
✓	✓	✓	×	93.2	87.1	90.9	59.8(+4.2)	2.13
✓	✓	✓	✓	93.9(+3.1)	88.6(+8.2)	91.7(+4.6)	60.8(+5.2)	2.76

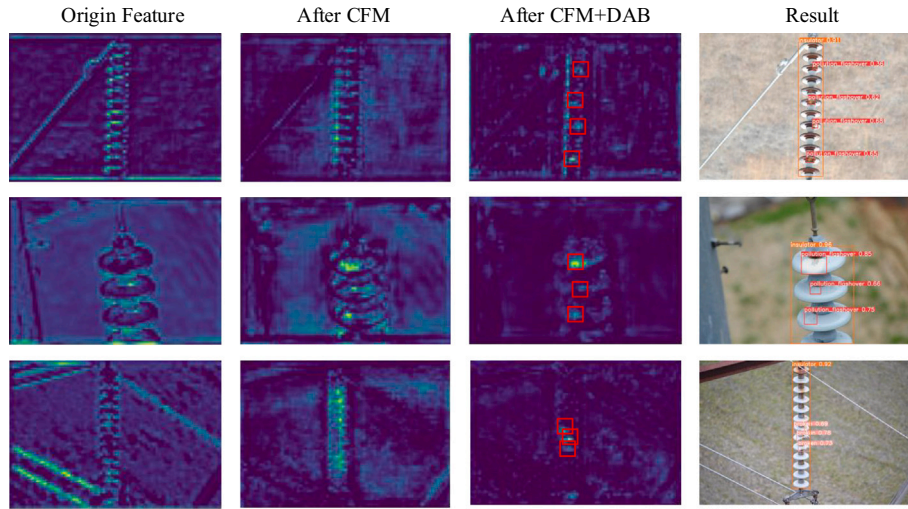


Fig. 7. Influence of CFM and DAB on feature extraction.

Flashover. Compared to other lightweight models, MID-YOLO leads in performance across the board. Specifically, against the FNet model, which is also optimized for insulator detection, MID-YOLO offers comparable FPS but leads by 0.4 in mAP50. Compared with the MFI-YOLO model, MID-YOLO shows better detection performance in insulator breakage and flashover, and also achieves comparable FPS with MFI-YOLO. Even against the larger YOLOv8s model, MID-YOLO achieves comparable performance. While SparseRCNN achieves the best performance, the RCNN family of models is overly large. Compared to MID-YOLO, it increases mAP50 by 0.06 but at a speed ten times slower. This underscores the efficiency and practicality of MID-YOLO's structural and modular design.

4. Conclusion

Building upon the YOLOv8 framework, this paper presents a novel algorithm tailored for the localization of various types of insulator defects. An analysis of the characteristics of these defects reveals that their detection is impeded by the complexity of natural backgrounds and the small size of the defects themselves. To address these challenges, we introduced an innovative neck feature fusion structure and a detail-aware module preceding the detection head. These enhancements significantly improve the model's ability to recognize small targets in complex scenes, resulting in a 3.1% increase in mean accuracy and an 8.2% improvement in recall rate. Specifically,

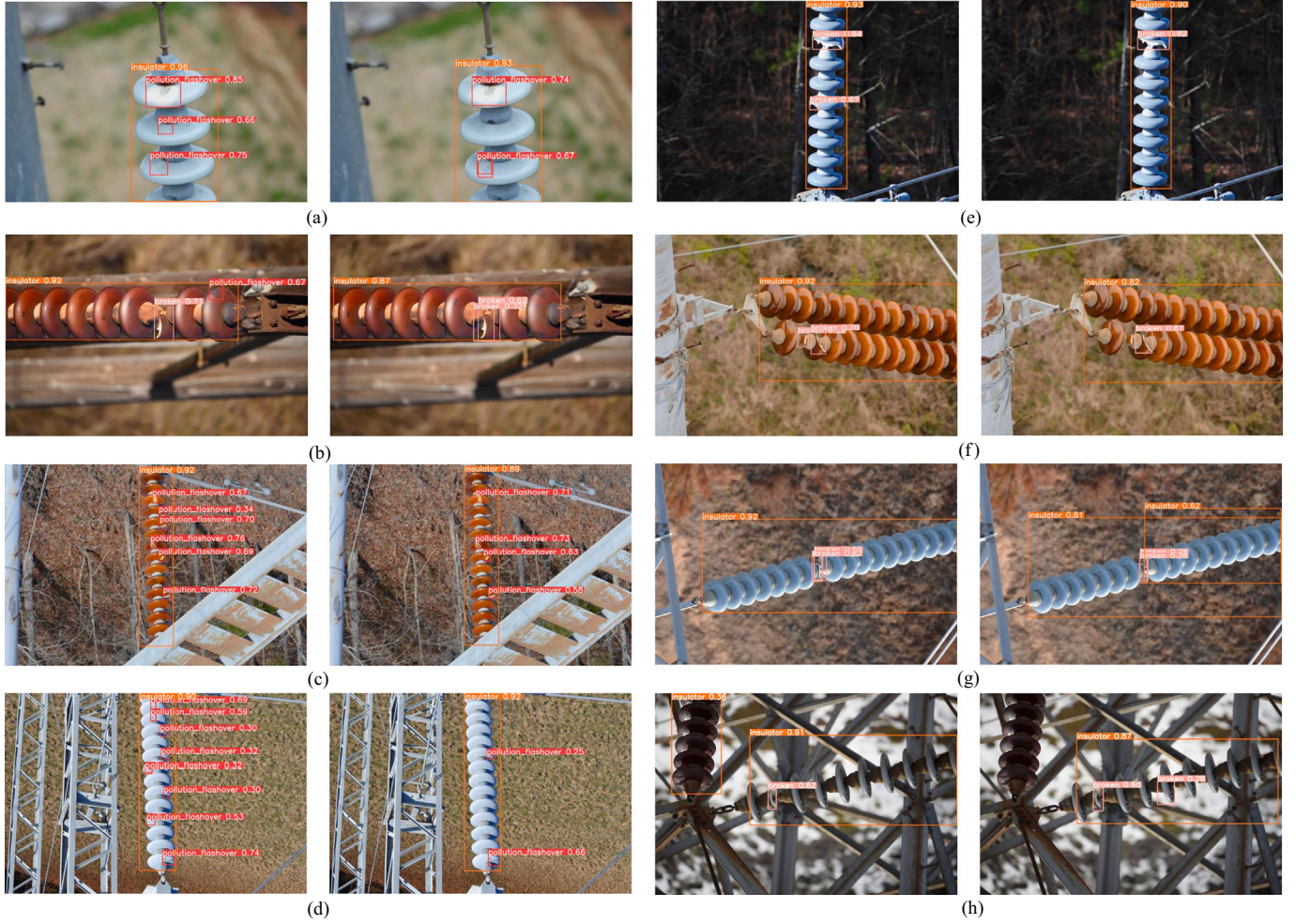


Fig. 8. Comparison of detection performance between MID-YOLO and YOLOv8 under various conditions, with MID-YOLO results on the left and YOLOv8 results on the right.

Table 3
Comparison with previous methods.

Models	Flashover	Broken	Insulator	mAP50	mAP50:95	Para(M)	FPS
FastRCNN	0.702	0.744	0.964	0.803	–	322	4.1
YOLOv5n	0.739	0.843	0.995	0.859	0.550	2.50	212
YOLOv8n	0.762	0.849	0.995	0.869	0.556	2.15	188
FINet	0.78	0.836	0.995	0.87	0.565	2.57	185
YOLOv8s	0.839	0.913	0.995	0.915	0.609	7.74	166
MFI-YOLO	0.840	0.917	0.995	0.916	0.608	2.73	183
Sparses RCNN	0.84	0.934	0.995	0.923	–	124	18
MID-YOLO	0.830	0.926	0.995	0.917	0.608	2.76	182

- (1) The CFM module, through its efficient fusion structure and the strategy of weighted channel processing provided by the TCB, effectively utilizes features across different channels, thereby enhancing the network's feature extraction capability and increasing the recall rate by 2.7%.
- (2) The MRB neck feature extraction block abandons the original, more complex, and redundant deep feature extraction strategy in favor of a multi-branch, low-dimensional breadth feature extraction strategy. This not only boosts the recall rate by 6.7% but also reduces the number of parameters by 7%.
- (3) The DAB module, considering the interconnection between spatial and channel features, introduces a synergistic attention mechanism to enhance the network's detail perception capability. It also significantly reduces the computational overhead

associated with the attention mechanism by rearranging and grouping features, leading to an all-around improvement in model metrics.

These distinctive designs allow the MID-YOLO to perform on par with models having a larger parameter count, such as YOLOv8s and Sparse RCNN, rendering it an efficient and practical solution for detecting insulator defects.

Although existing detection methods can identify three types of targets, numerous fault types remain unaddressed in real-world scenarios, such as flashover faults caused by rainfall or lightning strikes. To enhance the model's applicability in engineering practice, subsequent research should focus on constructing a more diverse defect sample

database. Consider integrating multi-modal data such as infrared thermal imaging, ultrasonic detection, and visible light to comprehensively assess insulator status across different physical characteristics, thereby enhancing the model's overall judgment capability under complex operating conditions.

Although this paper has optimized algorithms for small target recognition and complex background interference, practical deployment in power inspection, such as automated drone inspections, defect early warning and condition assessment systems, and maintenance priority determination, requires further evaluation of the model's generalization performance and engineering applicability through system integration and field validation.

CRedit authorship contribution statement

Lin Zhu: Writing – original draft, Methodology, Formal analysis, Software, Conceptualization. **Yuxing Mao:** Writing – review & editing. **Xingji Huang:** Supervision, Project administration. **Chunxu Chen:** Software. **Wenchao Yang:** Data curation. **Bozheng Lei:** Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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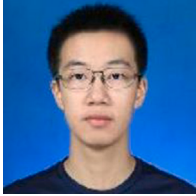
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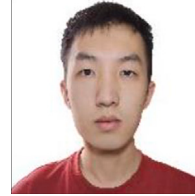
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